Does Education Raise Adult Income in China? Data Audit, Doubly Robust Evidence, and Weak-IV Diagnostics from CFPS 2022

Empirical Essay

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Abstract

This paper studies the relationship between education and adult income in China using CFPS 2022 microdata. Rather than relying only on a Mincer regression, the analysis first checks sample selection, missingness, wage outliers, and propensity-score overlap. It then treats college-or-above education as the main treatment and estimates its average treatment effect with cross-fitted augmented inverse probability weighting (AIPW). The paper also examines whether China's 1999 higher-education expansion generates a usable cohort-level instrument in this sample by reporting first-stage diagnostics. The final analytic sample contains 4,699 working-age adults. In the OLS benchmark, one additional year of schooling has a coefficient of 0.0882 on $\ln(\text{wage}+1)$, which equals a 9.22 percent difference in wage + 1 using the exact transformation $\exp(\beta) - 1$. College-or-above education has an OLS coefficient of 0.7405, or 109.7 percent after the same transformation. The preferred AIPW estimate is 0.7154, and alternative specifications range from 0.6859 to 0.7730. After weighting, standardized differences in age, gender, hukou status, urban residence, and marriage fall substantially. The expansion variable has a weak first stage in narrow cohort windows, so it is not used as the main causal evidence. Overall, the results show a stable positive education-income relationship; after balancing observed characteristics, college-or-above education remains strongly associated with higher wages.

Keywords: education, income, CFPS, China, AIPW, causal inference, higher-education expansion

1 Introduction

Education is one of the most important mechanisms through which individuals gain access to higher-paying jobs. Human-capital theory predicts that schooling raises productivity and therefore earnings [Becker, 1964, Mincer, 1974]. Yet estimating the return to education is difficult because schooling is not randomly assigned. Family background, ability, local labor-market conditions, and institutional barriers may jointly affect both education and income [Card, 1999, Heckman et al., 2008].

A regression of income on years of schooling can show an education-income gradient, but it leaves the analysis close to a descriptive correlation. This paper keeps the Mincer specification as a benchmark and then asks whether the income gap associated with college-or-above education remains after observed differences are balanced, and whether the 1999 higher-education expansion creates a sufficiently strong cohort-level shock to education in this CFPS cross-section. The treatment-effect question is addressed with doubly robust AIPW estimation [Hirano et al., 2003, Bang and Robins, 2005]; the policy-exposure question is evaluated through first-stage diagnostics around the 1981 birth-cohort cutoff.

This structure separates the descriptive result from the causal interpretation. OLS describes the education-income gradient, AIPW provides a treatment-effect estimate under selection on observables, and the expansion design is treated as a policy-based identification attempt that must pass a first-stage check before it can be interpreted as IV evidence.

2 Literature and Identification

The empirical starting point is the Mincer earnings function [Mincer, 1974]. A large literature finds that education is positively associated with earnings, but also stresses that OLS can be biased by unobserved ability or background [Card, 1999]. For China, Zhang et al. [2005] document rising returns to schooling during the market transition, while Asadullah and Xiao [2020] and Ma and Iwasaki [2021] show that returns vary across time, gender, hukou, region, and sector.

Recent work on China's higher-education expansion is especially relevant. Huang et al. [2022] use the great expansion to study returns to education and hukou-related differences. This paper borrows that policy intuition but first checks whether the available CFPS 2022 cross-section supports a strong first stage. Broad cohort windows show a visible education difference, but narrow local windows do not. Following standard RD guidance [Imbens and Lemieux, 2008, Lee and Lemieux, 2010], the policy design is therefore kept as a diagnostic rather than treated as the main source of identification.

The main causal framework is selection on observables. AIPW combines an outcome model and a propensity-score model, and remains consistent if either nuisance model is correctly specified under conditional ignorability [Bang and Robins, 2005]. Because this assumption cannot be proven from the data alone, the paper reports overlap and balance diagnostics and avoids reading the estimate as equivalent to a randomized or strong-IV result.

3 Data and Audit

The data come from CFPS 2022, released by the Institute of Social Science Survey at Peking University [Xie and Lu, 2015, Institute of Social Science Survey, Peking University, 2022]. The initial file has 9,806 observations. To keep the income outcome comparable across respondents, the analytic sample is limited to adults aged 18–64 with observed wage income and complete core controls. Table 1 reports the sample flow.

Table 1: Sample Construction

Step	N	Rule
Raw CFPS 2022 data file	9,806	
Keep adults aged 18–64	7,975	Age restriction
Require wage variable observed	4,718	Outcome available
Require core variables complete	4,699	Final analytic sample

The main outcome is ln(wage+1). The continuous education variable is years of schooling. The AIPW treatment variable equals one if the respondent completed college or above. Table 2 shows that wage availability is the main source of sample loss, while most covariates have limited missingness. The wage distribution has a long right tail, with a 99th percentile of 300,000 yuan, so a winsorized-wage robustness check is included.

Table 2: Missingness Audit in the Raw Data

Variable	Missing N	Missing percent
wage	4,934	50.32
educ	1	0.01
age_	0	0.00
gen	0	0.00
mar	261	2.66
rural	97	0.99
urban	79	0.81
health	139	1.42
inter	350	3.57
dw	408	4.16
asset	926	9.44

Table 3: Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
Annual wage income	4,699	60594.132	68985.459	0.000	48000.000	1800000.000
ln(wage + 1)	4,699	10.489	1.567	0.000	10.779	14.403
Years of education	4,699	11.006	4.376	0.000	12.000	22.000
College or above	4,699	0.347	0.476	0.000	0.000	1.000
Birth year	4,699	1981.133	10.921	1958.000	1982.000	2004.000
Age	4,699	40.867	10.921	18.000	40.000	64.000
Male	4,699	0.575	0.494	0.000	1.000	1.000
Rural hukou	4,699	0.795	0.404	0.000	1.000	1.000
Urban residence	4,667	0.637	0.481	0.000	1.000	1.000
Internet use	4,685	0.874	0.332	0.000	1.000	1.000
Self-rated health	4,673	3.215	1.045	1.000	3.000	5.000

Table 4: Education Distribution

Education category	N	Percent
Illiterate	243	5.17
Primary	589	12.53
Junior high	1,463	31.13
High school	774	16.47
College	702	14.94
Bachelor	824	17.54
Master	91	1.94
Doctorate	13	0.28

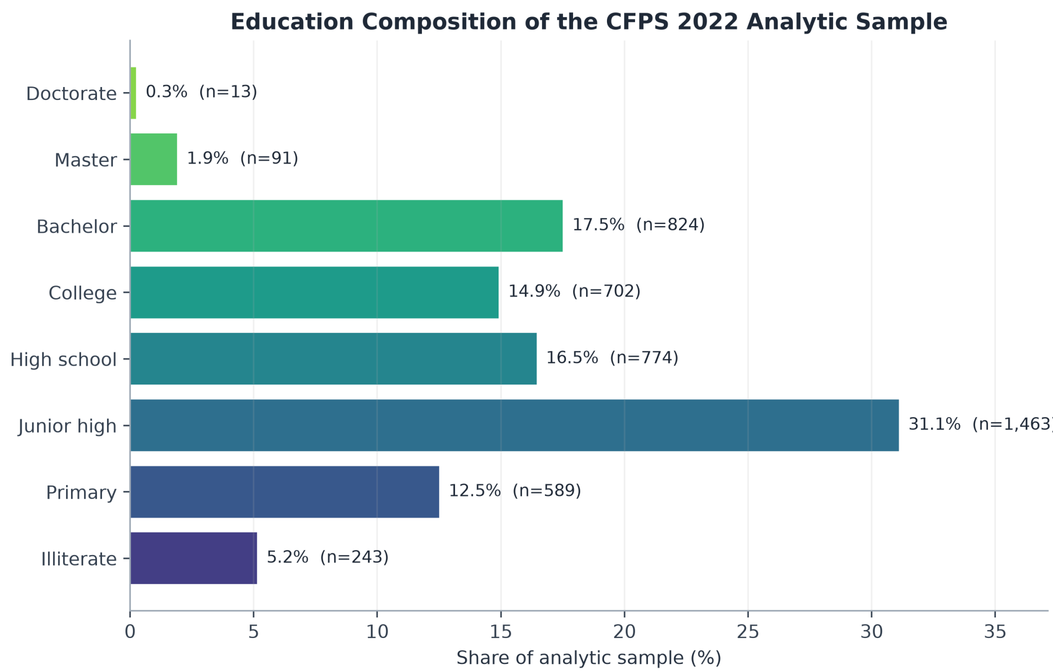

Figure 1: Education composition of the analytic sample

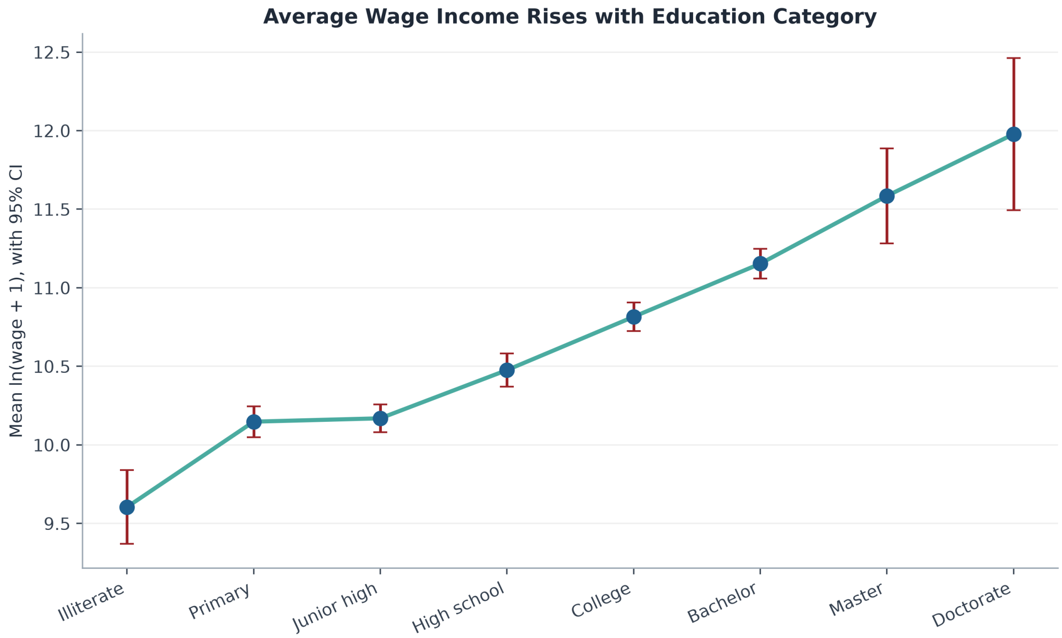

Figure 2: Mean log wage income by education category

Table 5 and Figure 3 make the slope between adjacent education levels explicit. The slope is not read as a log-point approximation. It is calculated as the change in mean $\ln(\text{wage}+1)$ divided by the change in mean years of schooling between neighboring education categories, and the final column reports the exact transformation $\exp(\Delta)-1$. The gradient is especially steep from high school to college, from college to bachelor, and from bachelor to master.

Table 5: Adjacent Income Slopes Across Education Levels

Education level	N	Mean years	Mean ln(wage+1)	Adjacent slope	Exact change (%)
Illiterate	243	0.00	9.604	–	–
Primary	589	6.00	10.148	0.0906	9.48
Junior high	1,463	9.00	10.169	0.0071	0.71
High school	774	12.00	10.476	0.1023	10.78
College	702	15.00	10.815	0.1131	11.97
Bachelor	824	16.00	11.153	0.3381	40.22
Master	91	19.00	11.585	0.1438	15.46
Doctorate	13	22.00	11.978	0.1311	14.01

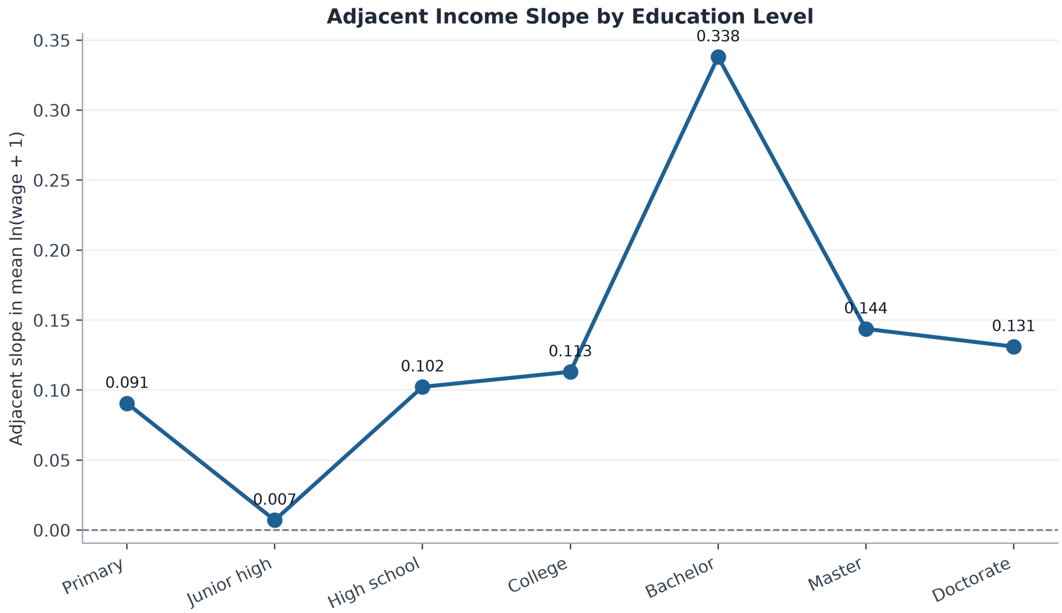

Figure 3: Adjacent education-income slopes with point labels

The working file only contains a binary `job` variable. It does not separate full-time work, part-time work, and non-employment, and it does not contain hours worked, so an hourly-wage outcome cannot be constructed without adding information that is not in the data. Table 6 reports the available employment-status measure, while the robustness checks include a positive-wage sample.

Table 6: Employment-Status Measure in the Working Data

Status	N	Percent	Mean wage	Mean ln(wage+1)
job = 0	265	5.64	33066.25	9.425
job = 1	4,434	94.36	62239.35	10.553

4 Empirical Strategy

4.1 OLS Benchmark

The benchmark model is

$$\ln(\text{wage}_i + 1) = \alpha + \beta E_i + \gamma X_i + \mu_p + \varepsilon_i, \tag{1}$$

where E_i is either years of schooling or college-or-above status, X_i includes demographic and household controls, and μ_p denotes province fixed effects. Robust standard errors are used throughout. These regressions provide the baseline education-income gradient, but the coefficients are not interpreted as fully causal.

4.2 Doubly Robust Treatment-Effect Estimation

The main estimand is the average treatment effect of college-or-above education:

$$ATE = E[Y_i(1) - Y_i(0)]. \tag{2}$$

The AIPW score is

$$\hat{\tau}_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}, \tag{3}$$

where D_i indicates college-or-above education, $\hat{e}(X_i)$ is the estimated propensity score, and $\hat{m}_d(X_i)$ is the predicted outcome under treatment status d. Five-fold cross-fitting is used so that the nuisance models are not trained and evaluated on exactly the same observations. The paper reports both linear and gradient-boosting nuisance models. The preferred specification uses age, gender, hukou, urban residence, marriage, and province. Extended specifications add health, household size, social status, internet use, and employment status as sensitivity controls.

4.3 Expansion-Based IV Diagnostic

China's higher-education expansion began in 1999. A natural idea is to use cohort exposure around the college-entry age as an instrument for education. The forcing variable is birth year minus 1981, because the 1981 birth cohort was approximately 18 in 1999. The diagnostic specification asks whether being born in or after 1981 creates a discontinuous increase in schooling within symmetric cohort windows. This design is useful only if the first stage is strong and local, so the first-stage coefficients and F-statistics are reported before any IV interpretation is considered.

5 Results

5.1 OLS Benchmarks

Table 7 reports the OLS benchmarks. One additional year of schooling has a coefficient of 0.0882 in the basic specification, which equals a 9.22 percent difference in wage + 1 after the exact transformation. With richer controls, the coefficient is 0.0824, or 8.59 percent. When education is measured as college-or-above completion, the coefficient is 0.7405 in the basic specification, or 109.7 percent after transformation; with richer controls, it is 0.6949, or 100.4 percent. The estimates remain positive across definitions of education and income, suggesting that the education-income gradient is not an artifact of one particular specification.

Table 7: OLS Benchmark Estimates

Specification	Coefficient	Robust SE	N	R^2
Years, basic controls	0.0882***	(0.0060)	4,699	0.158
Years, rich controls	0.0824***	(0.0063)	4,637	0.163
College, basic controls	0.7405***	(0.0488)	4,699	0.153
College, rich controls	0.6949***	(0.0497)	4,637	0.162
Winsorized wage	0.7317***	(0.0485)	4,699	0.152
Positive wage only	0.6438***	(0.0294)	4,638	0.285

Notes: Robust standard errors are in parentheses. Basic controls include age, age squared, gender, marriage, rural hukou, and province fixed effects. Rich controls additionally include urban residence, health, household size, internet use, and subjective social status. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

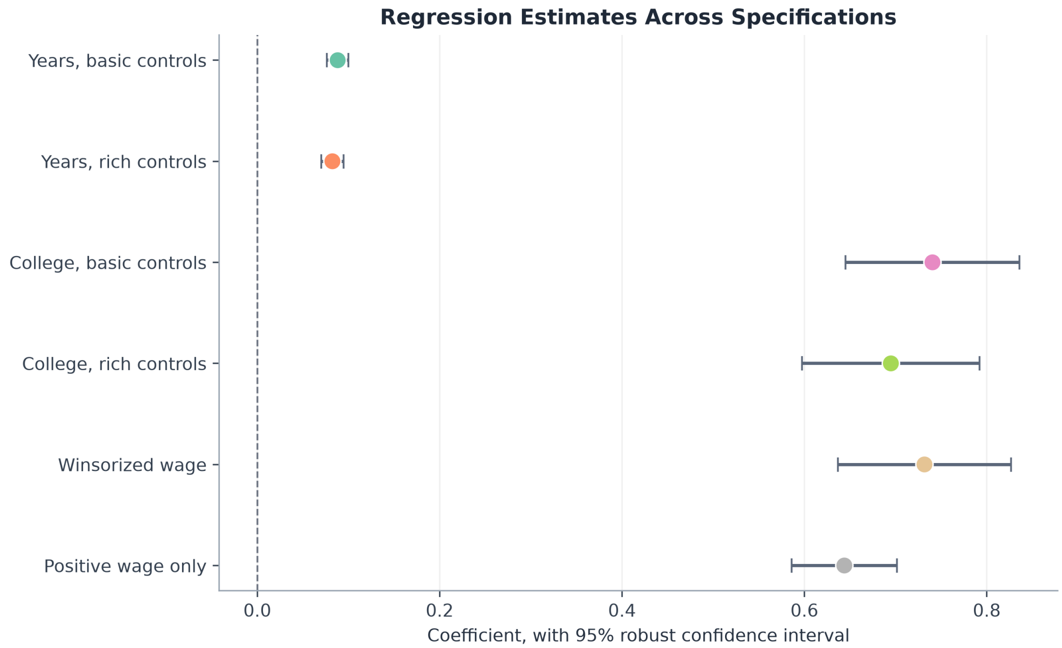

Figure 4: Regression coefficients across specifications

5.2 Doubly Robust Estimates

Table 8 reports the AIPW results. The preferred core-plus GBM estimate is 0.7154 with a standard error of 0.0570; the exact transformation gives a 104.5 percent difference in wage + 1 for college-or-above education. Across six specifications, estimates range from 0.6859 to 0.7730. These estimates are slightly smaller than the OLS college coefficient but still large, which suggests that the college-income premium remains visible after balancing observed differences between the two education groups.

Table 8: AIPW Estimates for College-or-Above Education

Covariates	Learner	ATE	SE	95% CI	N	AUC
Core	Linear	0.7726***	(0.0551)	[0.6647, 0.8806]	4,699	0.814
Core	GBM	0.7206***	(0.0553)	[0.6123, 0.8289]	4,699	0.822
Core plus	Linear	0.7730***	(0.0548)	[0.6655, 0.8804]	4,699	0.813
Core plus	GBM	0.7154***	(0.0570)	[0.6038, 0.8270]	4,699	0.824
Extended	Linear	0.7203***	(0.0517)	[0.6190, 0.8216]	4,699	0.827
Extended	GBM	0.6859***	(0.0489)	[0.5901, 0.7817]	4,699	0.843

Notes: The outcome is $\ln(\text{wage} + 1)$. ATE is the estimated average treatment effect of completing college or above. Linear uses logistic/Ridge nuisance models; GBM uses gradient boosting. The preferred specification is Core plus with GBM.

The credibility of AIPW depends on overlap and balance. Figure 5 shows that treatment and control units share a broad common propensity-score region, although a small number of observations lie near the tails. Those tails are a reminder that the estimate is not free of extrapolation risk. Table 9 and Figure 6 show that weighting reduces absolute standardized differences for core covariates to below 0.10.

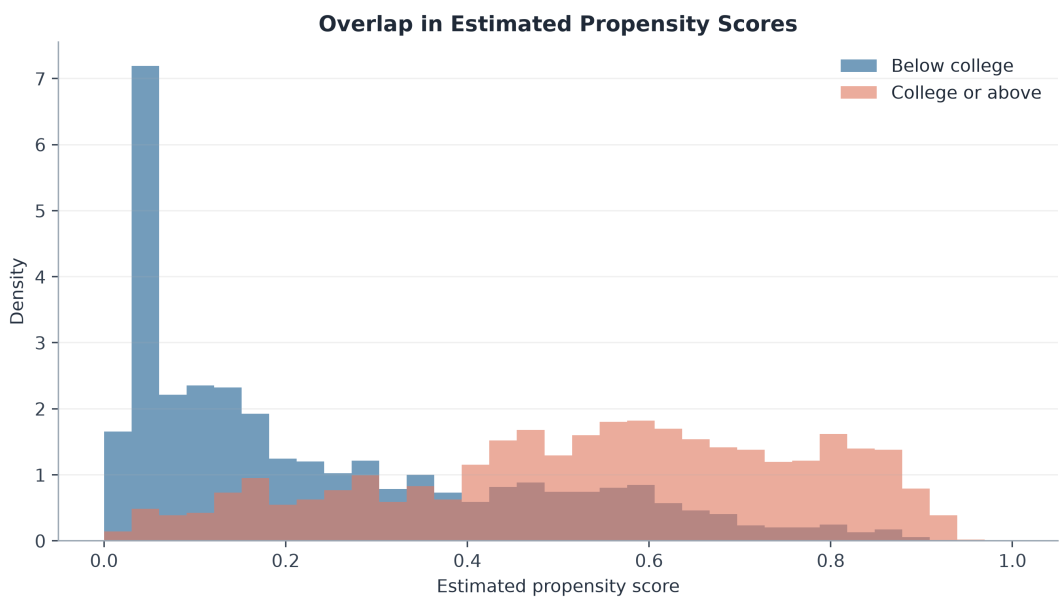

Figure 5: Propensity-score overlap

Table 9: Covariate Balance Before and After Weighting

Covariate	Raw SMD	Weighted SMD
age_	-0.937	-0.069
gen	-0.214	0.037
rural	-0.525	-0.021
urban	0.642	0.029
mar	-0.286	-0.026

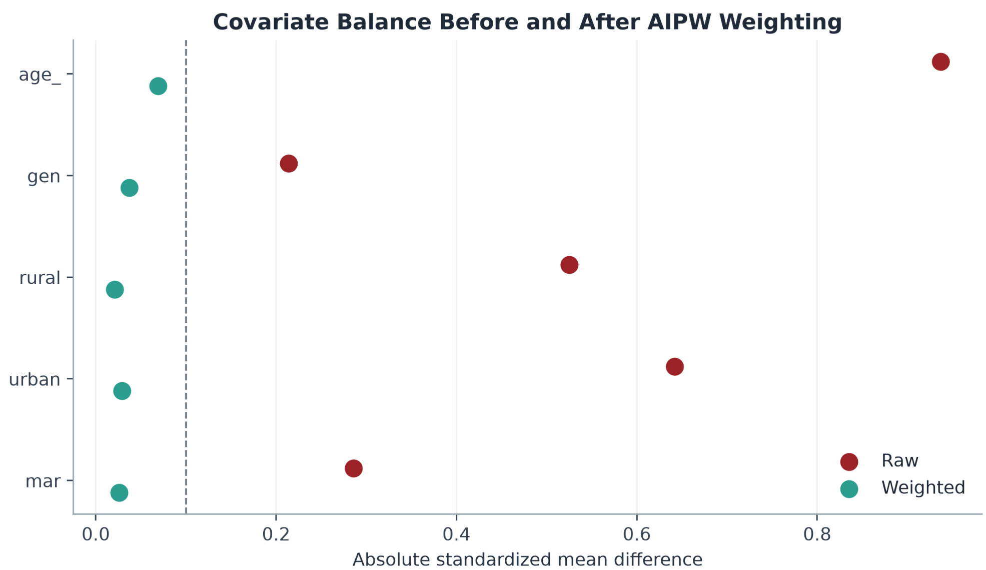

Figure 6: Standardized mean differences before and after weighting

5.3 Weak-IV Diagnostic

Figure 7 plots college completion and average schooling by birth cohort around the 1981 cutoff. Table 10 reports the formal first-stage and IV diagnostics. In narrow windows of ± 5 to ± 15 years, the first-stage F-statistics are far below conventional weak-instrument thresholds. The ± 20 window has a strong first stage, but it is too wide to read as a local discontinuity because it spans a broad range of cohorts and life-cycle positions. The policy-based IV design is therefore reported as a checked but unsuccessful identification route, not as the main causal estimate.

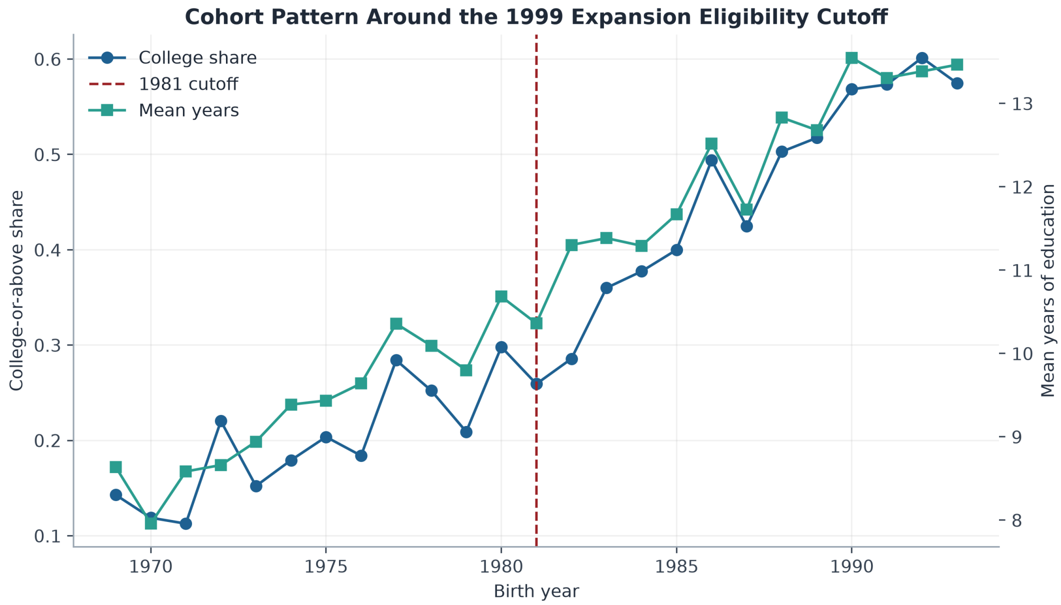

Figure 7: Cohort pattern around the expansion eligibility cutoff

Table 10: Higher-Education Expansion IV Diagnostics

Window	N	First stage	F-stat	IV estimate	SE
+/-5	1,280	-0.0344	0.01	1.1830	(14.9117)
+/-7	1,844	0.1990	0.29	0.4526	(0.9407)
+/-9	2,462	-0.0652	0.04	0.1819	(1.7998)
+/-12	3,277	0.1072	0.15	-0.9439	(2.8116)
+/-15	3,933	0.3806	2.30	-0.1649	(0.2887)
+/-20	4,596	1.2934	32.63	0.0411	(0.0636)

Notes: The instrument is an indicator for birth year 1981 or later, with linear running variable controls and province fixed effects. The weak first stage in narrow windows means the IV results should not be read as the main causal evidence.

5.4 Heterogeneity

Education-income associations are stronger for women than men and stronger for non-rural hukou holders than rural hukou holders. This is consistent with recent evidence that education returns in China vary by gender, hukou, sector, and region [Ma and Iwasaki, 2021, Li et al., 2025].

Table 11: Heterogeneous Education Coefficients

Subsample	Education coefficient	Robust SE	N	R^2
Women	0.1098***	(0.0091)	1,996	0.155
Men	0.0671***	(0.0067)	2,703	0.145
Non-rural hukou	0.1469***	(0.0198)	962	0.144
Rural hukou	0.0774***	(0.0064)	3,737	0.128

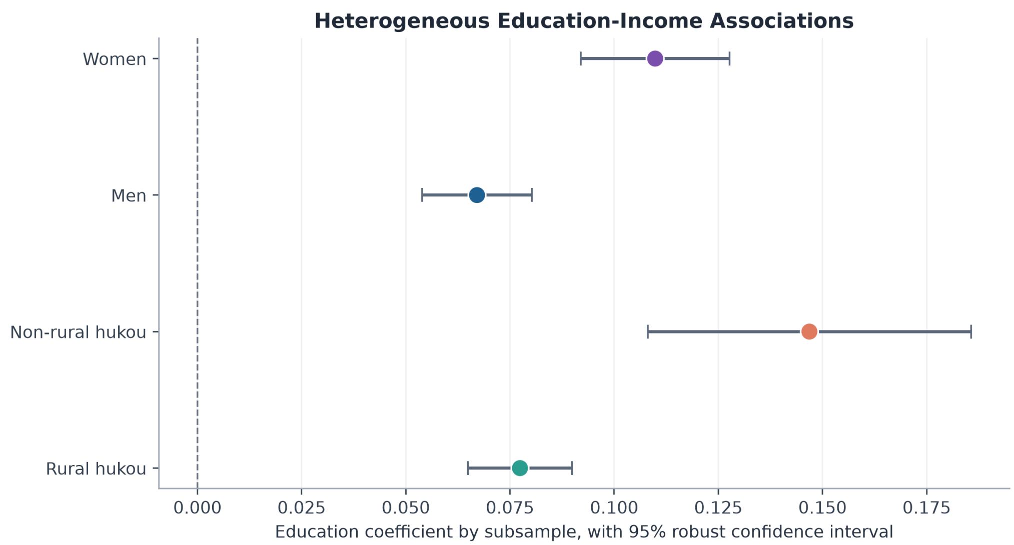

Figure 8: Education-income association by gender and hukou subsample

6 Discussion

The results should be read as a set of checks rather than as a single coefficient. The sample audit shows where observations are lost and how the wage tail is handled. The OLS models establish the baseline education-income gradient. The AIPW estimates show that the college-income gap remains large after balancing observed characteristics. The expansion diagnostic, meanwhile, shows that the current cross-section does not provide a strong local IV design. This combination gives a clearer empirical story than a standalone significant education coefficient.

The main limitation is that AIPW still depends on observed covariates. CFPS 2022 lacks direct measures of ability, parental education in the working file, school quality, and childhood family resources. Therefore, the preferred estimate should be interpreted as a selection-on-observables treatment-effect estimate, not as a definitive natural-experiment estimate. The weak-IV diagnostic also shows why it would be misleading to claim a strong expansion-based causal result from this cross-section alone.

7 Conclusion

Using CFPS 2022, this paper finds a large positive education-income relationship among Chinese adults. The conventional Mincer coefficient is 0.0882 per year of schooling, which corresponds to a 9.22 percent difference in wage+1 using $\exp(\beta)-1$. In the AIPW framework, the preferred estimate for completing college or above is 0.7154, or 104.5 percent after the same exact transformation. The estimate is stable across alternative nuisance models and covariate sets, and the balance diagnostics improve after weighting. A cautious reading is that, within the limits of the observed data, higher education remains closely linked to higher wages. Stronger causal claims would require a cleaner policy shock, panel evidence, or richer measures of family background and ability.

References

M. Niaz Asadullah and Saizi Xiao. The changing pattern of wage returns to education in post-reform china. *Structural Change and Economic Dynamics*, 53:137–148, 2020. doi: 10.1016/j.strueco.2020.01.010.

Heejung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005. doi: 10.1111/j.1541-0420.2005.00377.x.

Gary S. Becker. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. National Bureau of Economic Research, New York, 1964. URL https://www.nber.org/books-and-chapters/human-capital-theoretical-and-empirical-analysis-special-reference-education-first-edition.

David Card. The causal effect of education on earnings. In Orley C. Ashenfelter and David Card, editors, *Handbook of Labor Economics*, volume 3A, pages 1801–1863. Elsevier, 1999. doi: 10.1016/S1573-4463(99)03011-4.

James J. Heckman, Lance J. Lochner, and Petra E. Todd. Earnings functions and rates of return. *Journal of Human Capital*, 2(1):1–31, 2008. doi: 10.1086/587037.

Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

Bin Huang, Massimiliano Tani, Yi Wei, and Yu Zhu. Returns to education in china: Evidence from the great higher education expansion. *China Economic Review*, 74:101804, 2022. doi: 10.1016/j.chieco.2022.101804.

Guido W. Imbens and Thomas Lemieux. Regression discontinuity designs: A guide to practice. *Journal of Econometrics*, 142(2):615–635, 2008. doi: 10.1016/j.jeconom.2007.05.001.

Institute of Social Science Survey, Peking University. China family panel studies 2022. Survey data, 2022. URL https://www.isss.pku.edu.cn/cfps/.

David S. Lee and Thomas Lemieux. Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2):281–355, 2010. doi: 10.1257/jel.48.2.281.

Mingming Li, Xinyun Hu, and Keyan Jin. The return on education and the gender wage gap in china: A sector perspective. *SAGE Open*, 2025. doi: 10.1177/21582440251327015.

Xinxin Ma and Ichiro Iwasaki. Return to schooling in china: A large meta-analysis. *Education Economics*, 29(4):379–410, 2021. doi: 10.1080/09645292.2021.1900791.

Jacob Mincer. *Schooling, Experience, and Earnings*. National Bureau of Economic Research, New York, 1974. URL https://www.nber.org/books-and-chapters/schooling-experience-and-earnings.

Yu Xie and Ping Lu. The sampling design of the china family panel studies (cfps). *Chinese Journal of Sociology*, 1(4):471–484, 2015. doi: 10.1177/2057150X15614535.

Junsen Zhang, Yaohui Zhao, Albert Park, and Xiaoqing Song. Economic returns to schooling in urban china, 1988 to 2001. *Journal of Comparative Economics*, 33(4):730–752, 2005. doi: 10.1016/j.jce.2005.05.008.